

YULIANA ITUYAN ANGULO

Mechatronics Engineering Student | Software Analysis and Development Technologist

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RESEARCH PROFILE

Mechatronics Engineering student (expected graduation: October 2027) focused on computer vision and deep learning for embedded, resource-constrained systems. Author of a poster accepted at the Women in Machine Learning (WiML) Workshop at NeurIPS 2026. Currently conducting thesis research on an embedded computer vision system that relates facial behavioral patterns (attention, concentration, fatigue) to measured user performance and generates adaptive well-being feedback. Interested in developing accessible AI technology grounded in real-world needs, particularly in the Latin American context.

PUBLICATIONS

- Ituyan Angulo, Y. **Jazmín.IA: Embedded Computer Vision System for the Analysis of Behavioral Indicators and Their Relationship with User Performance.** Poster, Women in Machine Learning (WiML) Workshop at NeurIPS 2026.

EDUCATION

B.Eng. in Mechatronics Engineering 2020 – Oct 2027 (expected)

Universidad Autónoma de Occidente, Cali, Colombia

- Cumulative GPA: 3.7 / 5.0 (Colombian scale). GPA in programming and AI courses: above 4.0 / 5.0.
- Relevant coursework:** Neurocontrol (elective) – 4.6 / 5.0, covering the mathematical modeling of neural networks

Associate Degree in Software Analysis and Development 2023 – 2025

SENA (Servicio Nacional de Aprendizaje), Colombia

RESEARCH EXPERIENCE

Undergraduate Thesis – Embedded Computer Vision System Mar 2026 – Present

Universidad Autónoma de Occidente. Advisor: Javier Ferney Castillo

- Evaluating the relationship between facial behavioral patterns and user performance through an embedded computer vision system designed to support user well-being.
- Designing a video acquisition system on the Arduino UNO Q (4 GB) that captures facial images and processes all data locally to protect user privacy.
- Implementing computer vision algorithms (MediaPipe facial landmarks, Eye and Mouth Aspect Ratios, quantized TensorFlow Lite models) and microexpression recognition to extract facial patterns associated with attention, concentration, and fatigue.
- Measuring objective user performance in specific tasks (response time, accuracy, error rate) and analyzing its relationship with the facial patterns to generate adaptive feedback.
- Comparing the reliability and predictive utility of different versions of the AI models
- Results presented as a poster at the WiML Workshop, NeurIPS 2026.

Research Projects in Robotics 2024 – 2025

Universidad Autónoma de Occidente

- Autonomous unmanned aquatic robot for surface pool cleaning (team project): led the programming area, including robot teleoperation, the ROS implementation, and simulation in RViz and Gazebo to validate behavior before deployment. Also built an interface to visualize the robot's trajectory using an ESP32-CAM.
- Drone transport model for medical supply distribution in rural areas of Colombia: responsible for programming and control

TALKS AND PRESENTATIONS

Speaker, Jornada de Investigación – Universidad Autónoma de Occidente 2025

- Implementation of an unmanned autonomous aquatic robot for surface pool cleaning.

Speaker, Jornada de Investigación – Universidad Autónoma de Occidente 2024

- Development of a drone transport model for medical supply distribution in rural areas of Colombia.

HONORS AND AWARDS

Third Place, Rally Latinoamericano de Innovación, Universidad Autónoma de Occidente

Finalist, Todos a Salvo: Desafío Occidente (Channgé and Colmena Seguros)

Outstanding Teaching Performance Recognition, Kodland Latinoamérica *May 2026*

Awarded for managing the largest number of groups with excellent performance metrics.

Room Host and Speaker Support, II International Forum on Project Management (Management 5.0), Universidad Autónoma de Occidente *Jun 2025*

56 hours of communication, coordination and logistics.

TRAINING AND CERTIFICATIONS

- **9th Colombian HPC Summer School** (2026): parallel and distributed programming (OpenMP, MPI, C++, Python), LLMs and agents, quantum computing and foundation models.
- **Certified Scrum Professional – ScrumMaster (CSP-SM)**, CertiProf – Aug 2025
- **AI Immersion + Google Gemini**, Alura – Sep 2025
- **Bridge for Billions** incubation program – Nov 2025
- **Skills Development to Innovate in the Fourth Industrial Revolution (4RI)**, Universidad Nacional de Colombia – Nov 2020

PROFESSIONAL EXPERIENCE

Programming and Mathematics Tutor, Kodland Latinoamérica *Jul 2025 – Present*

- Teach programming (Scratch, Minecraft Education, Python) and mathematics to children from age 7 across Latin America.
- Guide students through technology projects and manage several groups at once while tracking individual progress.
- Communicate technical concepts clearly to learners of different levels.
- Received the company's outstanding teaching performance recognition (May 2026).

TECHNICAL SKILLS

AI / Machine Learning: Python, TensorFlow / Keras, TensorFlow Lite, Detectron2, CNNs, deep learning, computer vision, generative models, image processing.

Robotics / Embedded Systems: Arduino, ROS 2 Humble, Gazebo, RViz, sensors and actuators, autonomous systems.

Software: JavaScript, React, Angular, SQL, APIs, Git / GitHub.

LANGUAGES

Spanish (native); English (B2). TOEFL iBT: planned for 2026

RESEARCH INTERESTS

Computer vision, deep learning, embedded and resource-constrained AI, robotics, and AI applied to human well-being.